LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Video demonstrating fluid flow characteristics
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Video demonstrating fluid flow characteristics. Video on theory of flight
5. Specific Objectives	11.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review video segments; Instructor supervises note-taking on flow patterns and boundary layer separation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the basic properties of the atmosphere?
(b) Define the term boundary layer.
(c) How does fluid flow behave over a solid body?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Atmospheric properties and density • Viscosity and fluid behavior • Flow over bodies and streamlines • Bernoulli's principle application • Pressure distribution on surfaces	L & Ds	PPT, IB	15 min
(b)	• Boundary layer formation and growth • Laminar and turbulent flow characteristics • Separation points and wake formation • Effect of surface roughness • Reynolds number significance	L & Ds	PPT, PDF	20 min
(c)	• Flow visualization techniques • Impact of angle of attack • Drag generation mechanisms • Lift generation basics • Connection to helicopter flight	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the boundary layer affects lift generation.
(b) Describe the difference between laminar and turbulent flow.
(c) How does fluid flow change when separation occurs?

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of fluid flow discussed in this lesson apply to the rotor blades covered in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Factors affecting CL, Theories of Lift production, force and pitching moment, coefficient of lift, variation of co-efficient of lift with aspect ratio and sweep back and CL.	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min